Survival Analysis of DirtSlurper3100 Components

February 1, 2026

Abstract

This report analyzes the reliability of three critical components (battery, IR sensor, and impact sensor) of the DirtSlurper3100 robotic vacuum cleaner. Using survival analysis techniques on data from over 6,000 units collected between 2015-2019, we assess component lifetimes against the claims made by the manufacturers and identify whether having pets, a certain portion of carpeted area (reflected by a carpet score variable) or usage intensity affects failure rates. The results of the analysis show high reliability for the IR and Impact Sensors, whereas the Battery demonstrates faster wear-out under intensive usage. Accordingly, a differentiated, usage-dependent warranty policy is proposed to ensure fair and data-driven coverage.

1 Introduction

The DirtSlurper3100 is a robotic vacuum cleaner equipped with multiple sensors and a rechargeable battery system. Reliability is a key concern for the manufacturer, as component failures increase warranty repair costs and affect customer satisfaction. This study focuses on three critical components: the battery, which powers all systems and gradually degrades with use; the infrared (IR) sensor, which maps the environment and estimates distances for navigation; and the impact sensor, which detects collisions and protects the device from physical damage. Failures in any of these components compromise device functionality and may trigger costly warranty repairs.

The purpose of this project is to evaluate the reliability of these components using survival analysis. By examining their lifetimes and assessing the influence of household factors, such as pet ownership and flooring type, the study aims to identify the main drivers of failures. The results provide actionable insights for the manufacturer to optimize their warranty coverage.

This report is structured as follows: it begins with a characterization of the dataset, including preprocessing and exploratory data analysis; it then describes the methodology, covering the relevant models, hypothesis testing, and reliability assessment setup; this is followed by the results, and the report concludes with recommendations for the manufacturer.

2 The Dataset

The DirtSlurper3100 dataset contains above $6,000$ units registered after 2015 and monitored for operational status in 2019. The dataset captures registration date, total usage time, repair status, failure date, whether the owner has pets, a score to indicate the portion of carpeted area, and the condition of the three components: battery, impact sensor, and IR sensor.

2.1 Preprocessing

In this dataset, censoring was primarily administrative (i.e. devices that had not failed by December 31, 2019, were censored). This censoring is non-informative under the assumption that the study endpoint was predetermined and unrelated to device-specific failure tendencies.

Prior to survival analysis, the dataset was cleaned and reformatted to ensure internal consistency and proper handling of censored observations. Three devices marked as sent for repair but lacking failure dates were removed, while 64 devices with operational components but recorded failure dates were retained; these were included in the overall device-level analysis but treated as censored in component-level analyses since they likely failed due to an unobserved component. Records with zero usage time, typically corresponding to devices registered on the final study day, were excluded.

Categorical variables were recoded for modeling, including binary indicators for repair status and pet ownership, and numeric conversion of carpet score and total usage time. Component status variables for the battery, IR sensor, and impact sensor were encoded as 1 for failure and 0 for continued functionality.

Registration and failure dates were converted to valid date objects; missing failure dates were treated as right-censored by assigning the final study date (31 December 2019) and component status 0, allowing censored observations to contribute valid survival time without biasing failure rate estimates. A new variable, possession time, representing the duration between registration and failure or censoring in days, was created as the primary time-to-event input for survival modeling, with total usage time included as a covariate reflecting usage intensity.

These steps standardized and validated the dataset for survival analysis, ensuring consistent event definitions, correctly formatted variables, and accurate treatment of censored and uncensored observations.

2.2 Exploratory Data Analysis

To evaluate potential censoring, the distribution of repair status was examined. As shown in Figure 1, of the total units, 1,114 (17.2%) were sent to be repaired, while 5,359 (82.9%) were not, reflecting substantial right-censoring since most units were still functioning at the end of the study period.

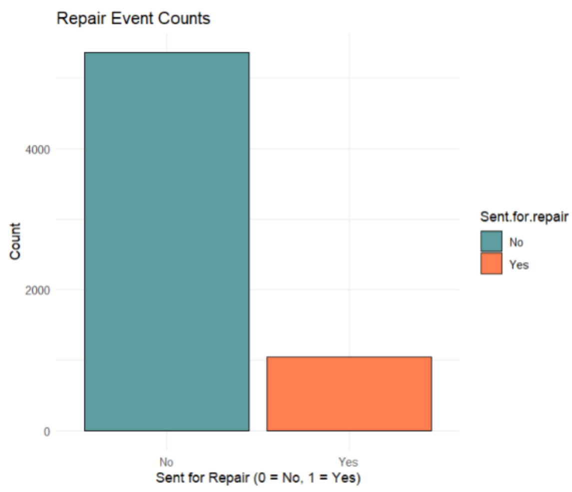

Figure 1: Distribution of the vacuum repair status.

Figure 2 shows the total usage time distribution by repair status. The maximum usage time is 3,819 hours and the median is 991 hours, revealing a right-skewed pattern, with fewer units used far more intensively than most.

Environmental covariates add additional insight which can help form expectations on the influence of these covariates on the survival of the device based on intuition. Pet ownership is reported in 2,629 households (40.6%), while 3,844 households (59.4%) reported no pets. Figure 3 shows that the average usage time is higher in homes with pets, as expected

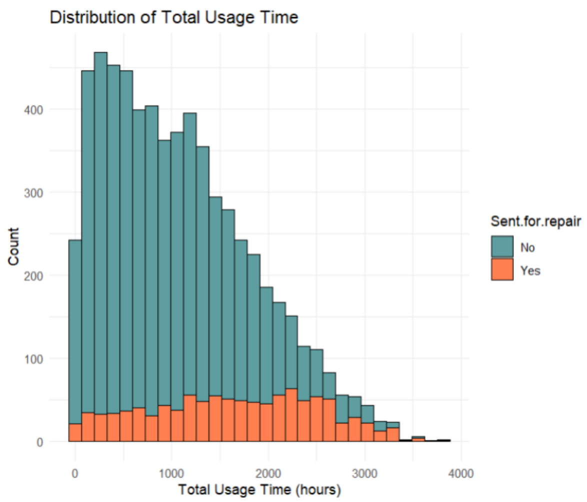

Figure 2: Bar chart showing the distribution of the total usage time of devices based on repair status.

due to greater cleaning needs, suggesting potential sensitivity of device components to pet presence. Carpet scores ranged from 1 to 8 (median = 3), with most between 2 and 4. Average usage time showed little variation across carpet scores, indicating minimal influence of flooring type on device use.

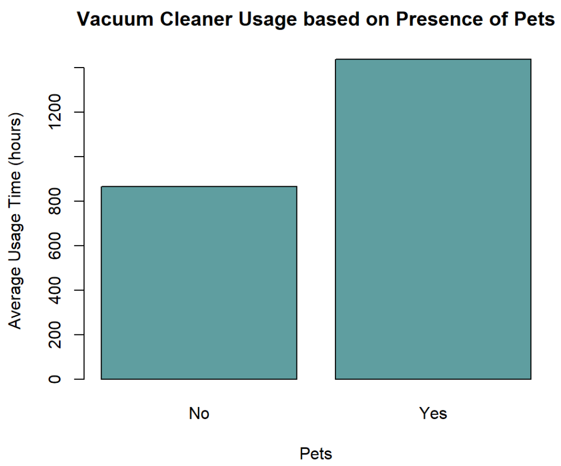

Figure 3: Bar chart showing average usage time by pet ownership. Vacuums in households with pets were used more intensively.

Examining possession time against total usage time reveals a clear positive correlation—devices owned for longer periods tend to accumulate more operating hours, as expected. Figure 4 illustrates this relationship, further distinguishing between devices that were repaired and those that were not. The trend lines for each group indicate that repaired devices generally experienced higher usage intensity, suggesting that higher operational demand may increase the likelihood of failure or the need for maintenance.

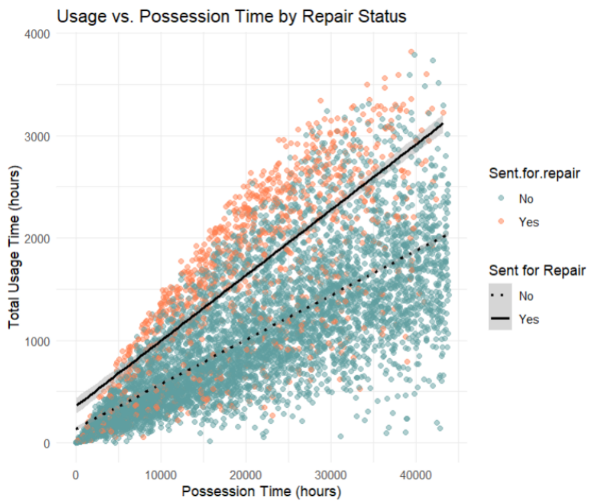

Figure 4: Relationship between possession time and total usage time, grouped by repair status.

Component status distributions provide additional insight (Figure 5). Among recorded failures, the battery is the most failure-prone, with 679 failures compared to 435 instances where it was not responsible, indicating that batteries were the primary cause of failure in the dataset. In contrast, only 91 failures were attributed to the impact sensor, suggesting that the dataset may underrepresent failures for this component. The IR sensor falls in between, with 281 failures attributed to its damage.

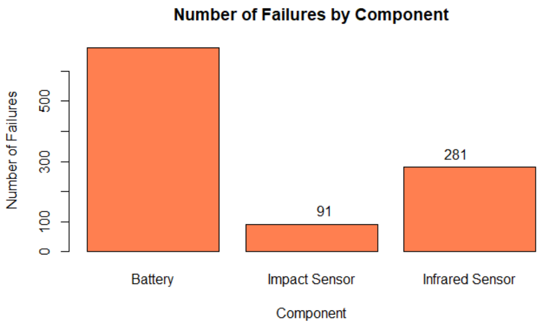

Figure 5: Number of failures observed for each component.

3 Methodology

To characterize the lifetime of each of the three components, we used a combination of non-parametric, parametric, and semi-parametric models. This multi-model framework provides complementary insights: Kaplan–Meier estimation offers distribution-free survival summaries, parametric models enable lifetime characterization and hypothesis testing, and Cox models capture covariate effects without requiring a specified baseline hazard.

3.1 Survival Modeling Framework

Non-parametric Model: Kaplan–Meier Estimator

The Kaplan–Meier estimator was used to estimate the survival function for each component without assuming a specific lifetime distribution. This approach effectively handles right-censored data, and provides a step-wise estimate of survival probability over time. Kaplan–Meier curves were computed for the vacuum, as well as each component, considering the covariates that indicated pet ownership, carpet coverage level, and usage intensity to explore differences in survival patterns across subgroups.

Parametric Model: Exponential and Weibull Distributions

Parametric accelerated failure time (AFT) models were fitted using exponential and Weibull distributions for the specific components using the same covariates.

The Weibull model was selected for its flexibility, allowing increasing, decreasing, or constant hazard rates through its shape parameter, thereby accommodating diverse failure mechanisms from early-life defects to wear-outs. The exponential distribution assumes a constant hazard rate over time and serves as a simpler, memory-less benchmark.

Both parameters were estimated using maximum likelihood, and covariates were modeled multiplicatively on hazard rates. This approach provides an intuitive interpretation, as coefficients quantify the proportional change in expected lifetime associated with changes in covariates. For instance, pet ownership and usage intensity plausibly accelerate component degradation through increased operational stress, while carpet score might affect the rate of wear through increased mechanical resistance. These coefficients were used just to support the covariates effects on survival rate.

Semi-parametric Model: Cox Proportional Hazards

To evaluate the influence of operational and environmental factors, we estimated Cox proportional hazards (PH) models. The Cox model does not assume a specific baseline hazard distribution, making it suitable for assessing covariate effects. The covariates to fit the model remain the same (pet ownership, carpet score and total usage time). The vacuum and component-specific models were estimated to isolate individual contributions here as well. Proportional hazards assumptions were verified using Cox–Snell residuals and cumulative hazard plots.

3.2 Model Selection

Model comparison between Weibull and exponential specifications was performed using likelihood ratio tests (LRTs) and Akaike Information Criterion (AIC) values.

The LRT formally tests whether the additional shape parameter in the Weibull model significantly improves the fit relative to the exponential model. The test statistic follows a chi-squared distribution with one degree of freedom under the null hypothesis that the exponential model is adequate.

The AIC criterion balances the model fit against complexity through a penalized likelihood measure, with lower values indicating preferable models.

The model selection results can be found in Section 4.1

3.3 Hypothesis testing & Reliability assessment

Verifying Component Lifetime Claims

To evaluate whether the components meet the manufacturer's reliability claims, formal hypotheses were formulated based on available specifications or reference criteria.

- The IR sensor manufacturer claims that at least 90% of the parts will survive at least 2000 days. *(Results in Section 4.2)*

- The battery manufacturer claims that at least 90% of the parts will survive at least 1000 days if not used intensively (under 2400 hours). *(Results in Section 4.3)*

- The impact sensor manufacturer provided no reliability specifications; therefore, the lifetime of the sensors is analyzed to determine whether at least 90% of units survive 2000 days, using the IR sensor's operating timeframe as the minimum reference period. *(Results in Section 4.4)*

Each hypothesis was evaluated using Kaplan–Meier survival estimates with Greenwood's variance to calculate standard errors, allowing assessment of whether the null hypothesis could be rejected in a one-sided test at the chosen significance level $\alpha = 0.05$.

Covariate Effects on Component Survival:

The influence of environmental and operational factors on a failing component's survival was assessed using Kaplan–Meier survival estimates. Variables such as Pets, Carpet score, and Usage time were analyzed to determine their impact on reliability. This analysis provides insights for warranty recommendations and identifies factors that may reduce component lifetimes, revealing why the manufacturer may be seeing more failures than expected. *(Results in Section 4.5)*

4 Results and Inference

This section presents the results of the model selection process, followed by hypothesis testing and further investigation of covariate effects. These analyses form the basis for the inferences and recommendations regarding the appropriate warranty period.

4.1 Model Selection Results

Tables 1 and 2 summarize the parameter estimates and model likelihoods for the Weibull and Exponential AFT specifications, respectively. Model comparison statistics are reported in Table 3.

For the battery, the Weibull model provided a substantially better fit (Log-likelihood = -

Table 1: Weibull Model Results for Each Component

Component	Intercept	Total Usage Time	Pets	Carpet Score	Scale	Log-Likelihood
Battery	11.6	1.69×10^{-4}	-1.18	-0.0538	0.389	-8204.3
IR Sensor	10.4	1.64×10^{-3}	-0.76	0.0136	0.616	-3709.8
Impact Sensor	NA	NA	NA	NA	0.156	-57974

Table 2: Exponential Model Results for Each Component

Component	Intercept	Total Usage Time	Pets	Carpet Score	Log-Likelihood
Battery	14.4	-1.61×10^{-4}	-2.37	-0.134	-8514.0
IR Sensor	11.13	2.08×10^{-3}	-0.894	0.0167	-3755.3
Impact Sensor	16.82	-1.99×10^{-3}	-0.584	0.675	-1237.1

Table 3: AIC Comparison Between Weibull and Exponential Models

Component	df (Weibull/AIC)	Weibull AIC Score	Exponential AIC Score
Battery	5/4	16418.63	17035.94
IR Sensor	5/4	7429.64	7518.68
Impact Sensor	5/4	115958.02	2482.23

8,204.3; $p < 2 \times 10^{-16}$) than the exponential model, confirming a non-constant hazard pattern. The estimated shape parameter ($\approx 2.57 (> 1)$) indicates an increasing hazard rate; i.e. failures become more likely with time, due to wear-out. The LRT statistic of 619.32 and the AIC difference of $\approx$ 617 further supports Weibull as the superior model.

All three covariates are highly significant, with pets having the strongest effect (coefficient of -1.18, indicating a 69% reduction in expected survival time). The negative coefficient for carpet score suggests that increased carpet coverage accelerates battery degradation. Total usage time of the vacuum appears to be an influencing factor but not a governing one, because the impact of this coefficient only compounds in the long-run.

For the IR sensor, both models fit reasonably well, but the Weibull marginally outperformed the exponential (AIC difference of 89.1). The LRT statistic (LRT = 91.04, $p < 2.2 \times 10^{-16}$) strongly favors the Weibull specification. The shape parameter (≈ 1.62) suggests an almost constant hazard rate, indicating that failures occur randomly over time with limited aging effects.

For the Impact sensor, the Weibull model failed to converge, yielding non-finite coefficients and an implausible log-likelihood (–57,974), likely due to the small number of observed failures and high right-censoring. With only 91 failures among 6,466 devices, the data lacks sufficient information to reliably estimate a flexible hazard shape across covariate patterns. The Exponential model was therefore adopted, as it provided stable and interpretable results. The much lower AIC score for the exponential model (2482.2 vs. 115958) further supports this decision.

Log-rank tests were used to formally assess differences in survival distributions across the covariates. This non-parametric test compares observed and expected failure counts under the null hypothesis of identical survival distributions, producing a chi-squared test statistic with degrees of freedom equal to the number of groups minus one.

Overall, the selected parametric distributions captured the essential hazard behavior. The Weibull model provided a superior fit for the Battery and IR Sensor components, indicating time-dependent failure behavior, while the Exponential model was retained for the Impact Sensor due to convergence issues and data sparsity.

4.2 Hypothesis test 1: IR sensor

H_0 : The manufacturer's claim is false, i.e., $L_{10} \leq 2000$ days; equivalently, $S(2000) \leq 0.90$.

H_1 : The manufacturer's claim is true, i.e., $L_{10} > 2000$ days; equivalently, $S(2000) > 0.90$.

A one-sided hypothesis test was conducted at $t = 2000$ days to evaluate the manufacturer's reliability claim for the IR sensor.

The Kaplan–Meier estimator (as observed in Figure 6) for the IR sensor gave the survival probabilities shown in Table 4. At $t = 2000$ days, the estimated survival probability was 92%, with a standard error of 0.0057. Given $\hat{S}(2000) = 0.921 > 0.90$ and the p-value $= 0.0001 < 0.05$), there is strong statistical evidence supporting the manufacturer's claim that at least 90% of IR sensors survive beyond 2000 days. Additionally, the estimated L_{10} (time by which 10% of sensors fail) is infinite, indicating that fewer than 10% of units experienced failure during the observation period. This suggests that the IR sensor demonstrates exceptional reliability within the observed time window. For the log rank tests, with pets as a covariate, the p-value is 0.78407, and for the carpet as a covariate, the p-value is 0.90955. Both p-values are above 0.05, indicating no significant effect of either pets or carpet on IR sensor survival.

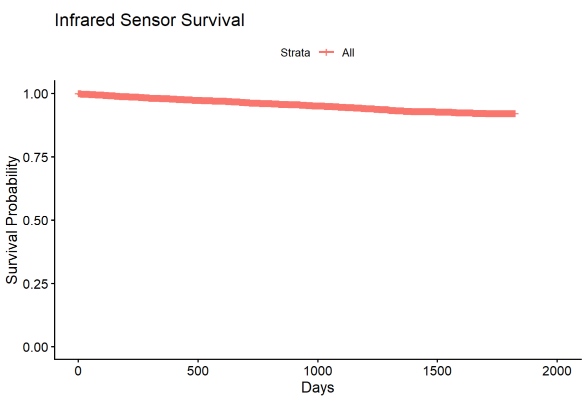

Figure 6: Kaplan–Meier survival curve for the IR sensor component.

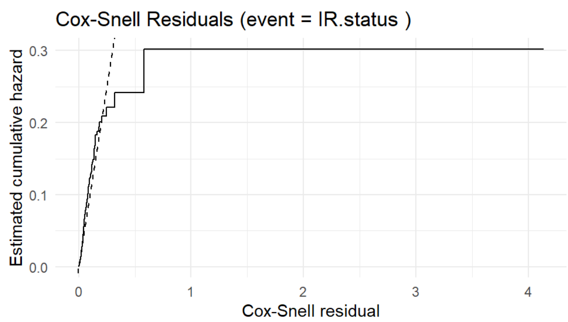

Figure 7: Cox–Snell residual plot for the IR sensor.

The Cox proportional hazards model provides further confirmation of these findings, as shown in Figure 7. Pet ownership is associated with a significantly higher hazard for the IR sensor, with a coefficient of 1.265, corresponding to a hazard ratio of approximately HR ≈ 3.54 (95% CI: 2.75–4.57, $p < 2 \times 10^{-16}$). In contrast, the carpet score has a negligible effect on the hazard (coefficient -0.022, HR ≈ 0.98, $p = 0.63$), indicating that carpet coverage does not meaningfully influence IR sensor failures. Total usage time has a very small negative association with the hazard (coefficient -0.0027, HR ≈ 0.997), suggesting a minimal effect of prolonged use. The model's concordance of 0.896 further indicates strong predictive accuracy for failure times. Overall, these results reinforce that the IR sensor is highly reliable under normal usage conditions, and that environmental factors such as pets or carpet coverage do not substantially affect its performance.

4.3 Hypothesis test 2: Battery

H_0 : The manufacturer's claim is false, i.e., $L_{10} \leq 1000$ days for batteries used < 2400 hours; equivalently, $S(1000) \leq 0.90$.

H_1 : The manufacturer's claim is true, i.e., $L_{10} > 1000$ days for batteries used < 2400 hours; equivalently, $S(1000) > 0.90$.

The corresponding Z-statistic for the test is given by:

$$Z = \frac{\hat{S}(2000) - 0.90}{SE(\hat{S}(2000))} = \frac{0.921 - 0.90}{0.0057} = 3.71$$

This results in a one-sided p-value of $p = 0.0001$.

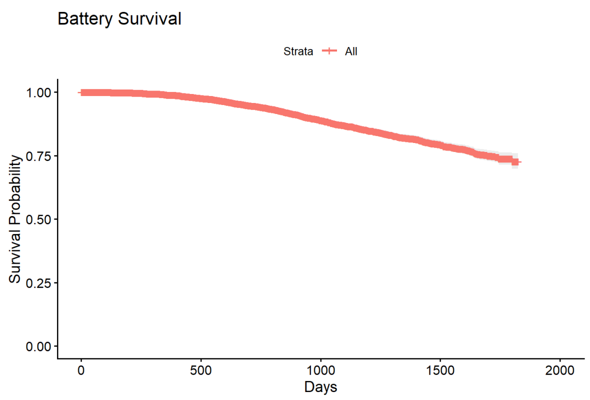

Figure 8: Kaplan–Meier survival curve for the Battery component, estimating survival probabilities over time.

To evaluate the manufacturer's reliability claim for the battery under regular usage conditions, a one-sided hypothesis test was performed on the Kaplan–Meier survival estimate (as observed in Figure 8) at $t = 1000$ days after filtering out devices that had a total usage time above 2400 hours. The Kaplan–Meier survival estimates for the battery (usage < 2400 hours) are summarized in Table 5.

At $t = 1000$ days, the estimated survival probability was 90.2%, which slightly exceeds the manufacturer's claim of 0.90, with a standard error of 0.0047. The estimated L_{10} is approximately 1013 days, consistent with the claimed performance threshold. However, the corresponding Z-statistic is 0.495, yielding a one-sided p-value of $p = 0.31$. This indicates that while the data supports the manufacturer's reliability claim, there is insufficient statistical evidence to reject the null hypothesis. Therefore, there is still a possibility that their claim is false.

The log-rank test for battery usage in the extreme case ($\geq 2400h$) usage group has showed reduced survival compared to normal usage ($< 2400h$) with a log-rank chi-square of 60.5 and p-value $\approx 7.44 \times 10^{-15}$. Pet ownership affected the battery survival with a chi-square of 878 and p-value $< 2.22 \times 10^{-16}$, indicating that batteries in homes with pets fail more than expected. Battery survival differed across carpet coverage groups with a chi-square of 13.1 and p-value 0.00146, so it's significant.

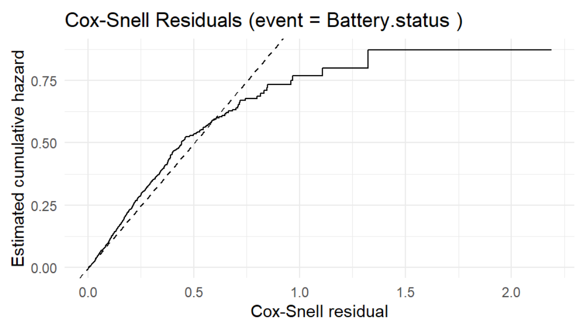

Figure 9: Cox–Snell residual plot for the battery component.

The results of the Cox proportional hazards model further support these findings for the battery, as illustrated in Figure 9. Pet ownership shows a very strong association with the hazard of battery failure (HR = 20.20, 95% CI: [15.78, 25.86], $p < 2 \times 10^{-16}$), indicating that batteries in households with pets are far more likely to fail. Carpet score also has a significant positive effect on failure risk (HR = 1.15,

Table 4: Estimated survival probabilities for the IR sensor.

Time (days)	**n.risk**	**n.event**	$\hat{S}(t)$	**Std. Err.**	**95% CI**
500	4524	144	0.974	0.0021	[0.970, 0.978]
1000	2757	81	0.953	0.0032	[0.946, 0.959]
1500	1097	51	0.928	0.0047	[0.919, 0.937]
1800	86	5	0.921	0.0057	[0.910, 0.932]

Table 5: Estimated survival probabilities for the battery (usage < 2400 hours).

Time (days)	**n.risk**	**n.event**	$\hat{S}(t)$	**Std. Err.**	**95% CI**
500	4539	121	0.976	0.0021	[0.972, 0.981]
1000	2604	279	0.902	0.0047	[0.893, 0.912]
1500	1004	96	0.857	0.0065	[0.844, 0.870]
1800	81	17	0.833	0.0088	[0.816, 0.851]

95% CI: [1.09, 1.21], $p = 8.33 \times 10^{-7}$), while total usage time shows a small but significant negative effect (HR = 0.9996, 95% CI: [0.9995, 0.9997], $p = 2.02 \times 10^{-12}$).

4.4 Hypothesis test 3: Impact

H_0 : $L_{10} \leq 2000$ days; equivalently, $S(2000) \leq 0.90$.

H_1 : $L_{10} > 2000$ days; equivalently, $S(2000) > 0.90$.

The Kaplan–Meier survival probabilities are summarized in Table 6.

At $t = 2000$ days, the estimated survival probability (as observed in Figure 10) is $\hat{S}(2000) = 0.930$ with a standard error of 0.0095.

The corresponding Z-statistic is 3.14, which gives a one-sided p-value of $p = 0.00085$.

The log-rank test comparing survival curves with covariate as pet ownership affected impact sensor survival, with a chi-square of 169 (df = 1) and p-value $< 2.22 \times 10^{-16}$, indicating higher failure rates in homes with pets. With covariate as carpet coverage also influenced impact sensor survival (chi-square = 29.8, df = 2, p = 3.46×10^{-7}), with failures more frequent in medium- and high-coverage areas.

Even though no official reliability claim was made, the analysis indicates that the impact sensor performs very well. The estimated survival at 1800 days is $\hat{S}(1800) \approx 0.93$, meaning at least 93% of units survive to this time. This suggests strong reliability of the impact sensor and supports its inclusion in warranty considerations.

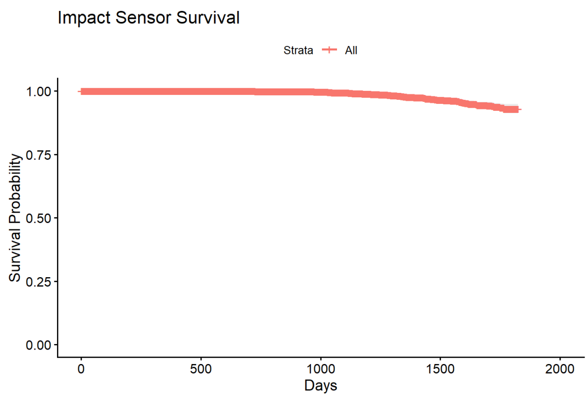

Figure 10: Kaplan–Meier survival curve for the Impact sensor component.

The results of the Cox model reinforce these findings, as illustrated in Figure 11. Pet ownership is associated with a significantly increased hazard, with a hazard ratio of about HR = 6.06, indicating that impact sensors in homes with pets are roughly six times more likely to fail compared to those without pets. In contrast, carpet score shows a negative association

Table 6: Kaplan–Meier survival estimates for the impact sensor.

Time (days)	**n.risk**	**n.event**	$\hat{S}(t)$	**Std. Err.**	**95% CI**
500	4650	0	1.000	0.0000	[1.000, 1.000]
1000	2889	10	0.997	0.0010	[0.995, 0.999]
1500	1139	60	0.965	0.0044	[0.956, 0.973]
1800	83	21	0.930	0.0095	[0.911, 0.949]

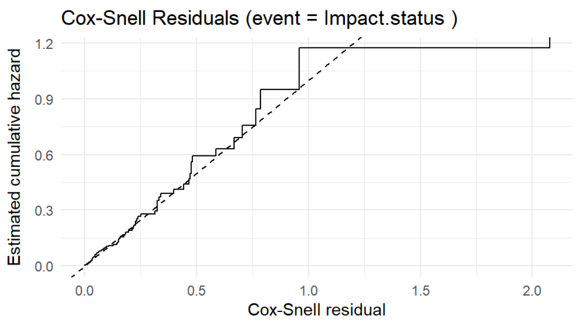

Figure 11: Cox–Snell residual plot for the impact sensor.

with the hazard (HR = 0.51), meaning higher carpet coverage is linked to a lower risk of impact sensor failure. Total usage time, though with a smaller effect, still contributes positively to the hazard, indicating that prolonged operation slightly increases the likelihood of failure. These results are consistent with the survival analysis trends and confirm that both environmental and usage factors shape the failure patterns of the impact sensor.

4.5 Findings on covariate influence

For the Battery and IR sensor components, the inclusion of covariates significantly improved model fit compared to their null counterparts without predictors. Specifically, the LRT statistics for the Battery ($\chi^2 = 947.84$, $p < 2.22 \times 10^{-16}$) and IR ($\chi^2 = 463.60$, $p < 2.22 \times 10^{-16}$) models strongly reject the null hypothesis (that covariates do not affect the failure time distribution), confirming that the covariates have a statistically significant effect on the failure time distribution. For the Impact sensor, the LRT ($\chi^2 = 113466.5$, $p < 2.22 \times 10^{-16}$) indicates that the covariates significantly affect failure times. The Weibull model shows a statistically better fit than the exponential model (LRT = 113473.8, $p < 2.22 \times 10^{-16}$), but convergence issues and unstable estimates due to heavy censoring (91 failures among 6,466 devices) make it unreliable. Hence, the exponential model was retained for interpretation, supported by its much lower AIC (2,482.2 vs. 115,958.0).

Overall, these results coincide with and validate the earlier findings presented in the covariate effect analysis: Total usage time, presence of pets, and the carpet score influence product longevity.

The Cox–Snell residuals for the full Cox model predict the vacuum cleaner failures and plot them against the estimated cumulative hazard, with a diagonal dashed reference line representing a perfect fit. The model demonstrates a good overall fit, as the estimated cumulative hazard, represented by the stepped line, closely follows the 45° reference line through a large portion of the plot and deviates only in certain regions. In the initial segment of the curve, up to approximately 1.5 residual units, the line remains very close to the reference line, indicating that the predicted hazard rates align well with the observed events during this period. When the stepped line deviates upward from the reference line, it suggests that the model is underestimating the hazard, meaning that more events are occurring than the model predicts. Conversely, when the curve deviates downward, it indicates that the model is overestimating the hazard, implying fewer observed events than expected. Towards the higher residual values, the curve flattens, reflecting fewer events or a small number of extreme observations in the tail of the distribution. This residual analysis provides meaningful insights into how long the vacuum cleaners last before requiring repair and illustrates how the considered covariates of pet ownership, carpet score, and total usage time capture the primary patterns in the failure data.

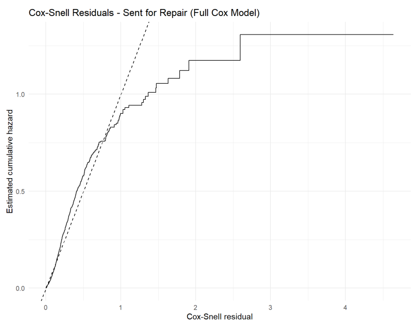

Figure 12: Cox Snell Residual for the Full Cox Model

The survival trends for different user environments appear broadly similar, with only minor deviations. This indicates that failures are occurring under normal operating conditions rather than being solely associated with harsh or extreme usage patterns (as observed in Figure 13). Therefore, management's concern about overall underperformance is valid, as failures cannot simply be attributed to a small subset of heavy users. Based on the updated analysis, batteries are likely the primary component contributing to early failures and that can be observed in Fig 14.

Although the survival probability for batteries remains relatively high (around 0.89 to 0.91), the L10 life is estimated at approximately 1013, and the p-value of 0.31 indicates low statistical confidence in the assumption that standard batteries will continue to per-

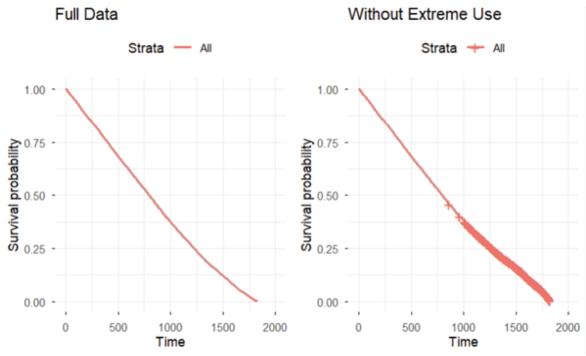

Figure 13: Comparing survival probabilities of all vacuums vs those that worked less than 2400 hours.

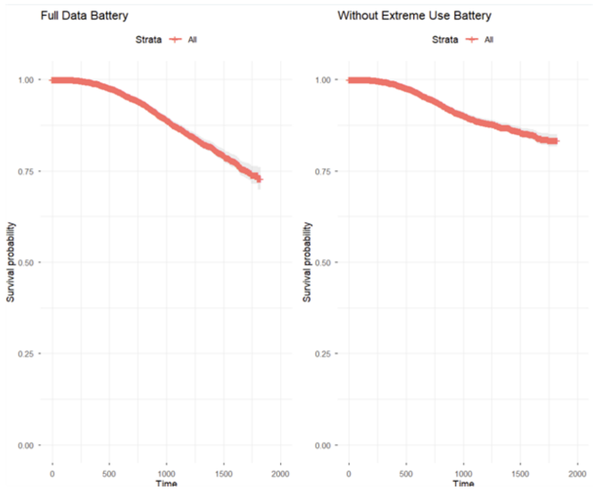

Figure 14: Comparing survival probabilities of all batteries vs those that worked less than 2400 hours.

form reliably. The battery is also observed to be one of the most vulnerable components of the vacuum in the analysis; therefore, the impact of total usage, presence of pets, and carpet area on its performance is studied to provide greater value.

While total usage does contribute to degradation, it is not the dominant cause of battery failure as seen in Figure 14. Additionally, environmental stressors such as the presence of pets (as observed in Figure 15) and extensive carpet coverage (as observed in Figure 16 emerge as some major causes of concern. These conditions introduce higher mechanical resistance, forcing the vacuum to draw more current, which accelerates battery wear even under moderate usage hours.

Therefore, battery failure is not caused randomly or solely by high operational time, but is instead attributed to consistent environmental stressors that standard usage metrics fail to capture. This accounts for the higher-than-expected warranty returns related to the vacuum, as traditional life prediction models assume uniform usage conditions and do not consider factors such as high-friction surfaces or pet-related particulate stress.

Overall, the data support the conclusion that pets and carpet area are major contributors to premature battery failures, while total usage has a long-term compounding impact on battery degradation. This, in turn, impacts the vacuum's overall lifespan and provides a strong explanation for the increased volume of warranty repairs observed.

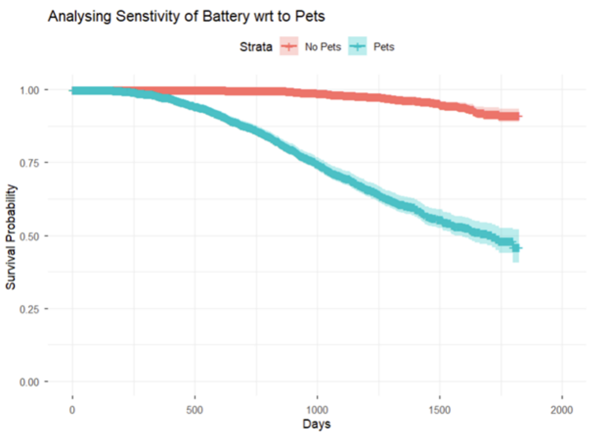

Figure 15: Survival rate of battery with respect to whether the household has pets.

5 Conclusions and Recommendations

This study combined parametric survival modeling, non-parametric hypothesis testing, and covariate analysis to assess the reliability of

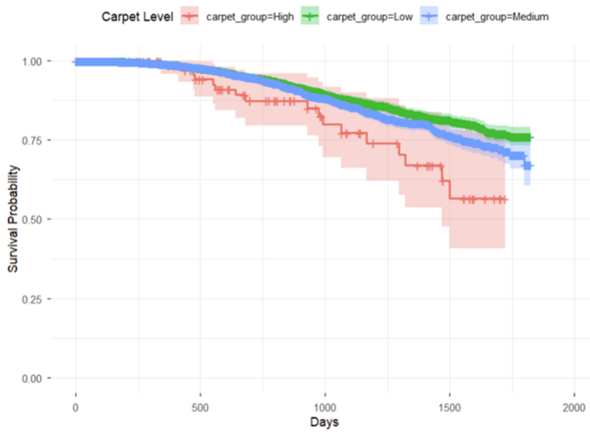

Figure 16: Survival rate of battery with respect to carpet score group (Low = score 1-3; Medium = score 4-6; High = 7-8).

key components in the robotic vacuum system. The results provide a statistically grounded basis for warranty determination and reliability improvement decisions.

The Battery and IR sensor components exhibited strong reliability characteristics under normal operating conditions, while the Impact sensor seemingly demonstrated outstanding performance with minimal recorded failures. However, this apparent robustness may reflect the limited number of observed failures rather than inherent durability; therefore, further investigation with additional data is recommended to confirm this finding.

Accordingly, the following component-specific recommendations are derived from the survival analysis:

IR Sensor. The IR sensor shows strong reliability, with an estimated survival probability of $S(2000) = 0.92$ and no significant covariate effects. A standard warranty period of up to five years (≈1825 days) is therefore statistically defensible.

Impact Sensor. Given the very low failure incidence ($S(1800) \approx 0.93$), the impact sensor can be safely covered under the same five-year warranty or used as a benchmark for system-level reliability.

Battery. The battery exhibits greater uncertainty, as the survival probability at 1000 days barely meets the 90% threshold and degrades more rapidly under heavy use. A two-tier warranty structure is therefore recommended to align coverage with operational stress:

- **Standard usage** (< 2400 hours): warranty coverage of approximately 2.5–3 years (900–1100 days).

- **Heavy usage** (≥ 2400 hours): warranty coverage of approximately 1.5–2 years, or exclusion from full warranty protection.

Implementing this approach would require reliable logging of device usage hours and clear communication with customers regarding the definition of usage tiers.

Since the battery is the most failure-prone component and highly sensitive to covariates, design improvements aimed at reducing wear-out effects are recommended. The observed decline in survival among devices operating in highly carpeted areas should also be investigated further, with input from domain experts to identify potential mechanical or environmental causes.

Overall, a general warranty period of five years is supported for the system, with the battery component governed by its own, usage-dependent policy. Continued data collection beyond five years will enable refinement of the survival models, ongoing reliability monitoring, and periodic revalidation of warranty assumptions using updated field data.

Appendix

A Code

The R scripts used for analysis are available in the following GitHub Project: https://github.com/NehaBogavarapu/Survival-Analysis-DirtSlurper3100-GA.git